BM3122 Medical Imaging

Intake	2020 onwards	**Specialization**	Biomedical Engineering		
Semester	**Code**	**Module Title**	**Credits**	**C/E/O**	**GPA/NGPA**
5	BM3122	Medical Imaging	3	C	GPA
Hours/Week		**Prerequisites and corequisites**		**Evaluation**	**%**
Lecture	**Lab./Tut.**			**CA**	**WE**
2	2	None		50	50

Module Aim: To provide the fundamental principles of medical image formation and how they interact with the human body.

Learning Outcomes

At the end of the module the student will be able to:

LO1	Explain fundamentals of image formation using signals obtained from the body
LO2	Compare and contrast characteristics of different medical imaging modalities
LO3	Identify effects of different imaging modalities on the human body
LO4	Interpret parameters of medical images for measurements and analysis

Outline Syllabus

1.	**Introduction to medical imaging [6 hrs]** Overview and examples, imaging signals and systems, image quality	LO1, LO2, LO3, LO4
2.	**Radiographic imaging [6 hrs]** Physics of radiography, projection radiography (X-ray), computed tomography (CT)	LO1, LO2, LO3, LO4
3.	**Magnetic Resonance Imaging (MRI) [8 hrs]** Nuclear magnetic resonance (NMR), Magnetic Resonance Imaging (MRI), functional MRI	LO1, LO2, LO3, LO4
4.	**Ultrasound imaging [3 hrs]** Ultrasound imaging principles, ultrasound imaging systems, Doppler ultrasound	LO1, LO2, LO3, LO4
5.	**Nuclear medicine imaging [3 hrs]** Radiopharmaceuticals, gamma camera, planar scintigraphy, single photon emission computed tomography (SPECT), positron-emission tomography (PET)	LO1, LO2, LO3, LO4
6.	**Optical and Thermal imaging [2 hrs]** Medical thermography, and Optical Coherent Tomography (OCT)	LO1, LO2, LO3, LO4

Mapping of Learning Outcomes to Program Outcomes and Assessment Methods															
LOs Covered	Assessments	PO-01	PO-02	PO-03	PO-04	PO-05	PO-06	PO-07	PO-08	PO-09	PO-10	PO-11	PO-12	PO-13	PO-14
LO1	A1, A2, A3, A4	H			L				L					L	
LO2	A1, A2, A3, A4	M			L			L	L		L			H	
LO3	A1, A3, A4							M	H	M		L		M	
LO4	A1, A3, A4				H			L				M		M	
Overall Contribution to POs		**H**			**H**			**M**	**H**	**M**	**L**	**M**			

Details on Assessment Methods

No.	Assessment Activity	%	LOs Covered	Comments
A1	Labs/Assignments – QA and Modalities	20%	LO-1, LO-2, LO-4	
A2	Presentation – QA and Modalities	10%	LO-1, LO-2, LO-4	
A3	Quizzes – In-class	20%	All	
A4	End Semester Examination	50%	All	

Recommended Textbooks	1. Jerry L. Prince, Jonathan Links, Medical Imaging Signals and Systems, Pearson Education, 2nd edition 2. M A Flower, Webb's Physics of Medical Imaging, CRC Press, 2nd Edition
Lecturer in Charge	Dr. Rukshani Liyanaarachchi